

UMA006: INTRODUCTORY MATHEMATICS-II

L	T	P	Cr
3	1	0	3.5

Course Objective: The objective is to develop basic computing skills and application of quantitative required for biological studies and rationalization of experimental designs.

Syllabus

Differentiation: Functions, Domain and range, Properties of standard functions (trigonometric, exponential and logarithmic) and their graphs, Limit, Continuity and Differentiability. Differentiation of standard functions (polynomials, trigonometric, inverse trigonometric exponentials and logarithmic), Product rule, Quotient rule, Chain rule, Applications of derivatives in graphing, Maximum and minimum of single variable function, Functions of several variables, Partial derivatives, Homogeneous functions, Maximum and minimum of several variable functions.

Integration: Integral as anti-derivative, Integration: by substitution, by parts and partial fractions, Definite integral and its properties, Double integrals, Areas of bounded regions and rectification.

Differential Equations: Order and degree, General and particular solution of differential equation, Techniques for solving first order ordinary differential equation and its applications to biological problems (population growth, radioactive decay).

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. Explain functions, related properties and determine their continuity and differentiability.
2. Apply derivatives in graphing and maxima and minima of single variable function.
3. Predict integration of function using by parts, by substitution and partial fraction methods and apply these to find areas of bounded regions and rectifications.
4. Learn methods to solve first order ordinary differential equations and apply it to biological problems.

Text Books

1. Mathematics, A Text books (Parts I & II), NCERT, New Delhi, 2011.
2. Thomas, G.B. and Finney, R.L. Calculus and Analytical Geometry, Pearson Education, 10th ed., 2007.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Reference Books

1. Kreyszig, Erwin, Advanced Engineering Mathematics, 8th Edition, John Wiley, 1999.
2. Shanti Narayan, Differential and Integral Calculus, S. Chand, 2005.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/tutorials/quizzes)	30

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